Skin cancer Detection using Hailo Computation power

After generating a .hef file we can compile the in two ways

➔ python hailo_apps/python/pipeline_apps/detection/detection.py \

--labels-json resources/json/barcode_labels.json \

--hef-path resources/models/hailo8l/yolov8s-hailo8l-barcode.hef \

--input resources/videos/barcode.mp4

This code will give the detection for the recorded video.

But our goal is live detection.

Use the following command for live detection

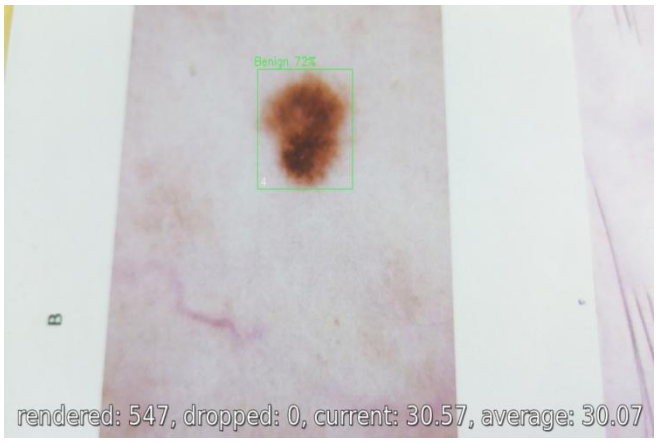
➔ hailo-detect --hef-path /home/satya/heffile/yolov8s.hef --input rpi --labels-json /home/satya/heffile/labels.json

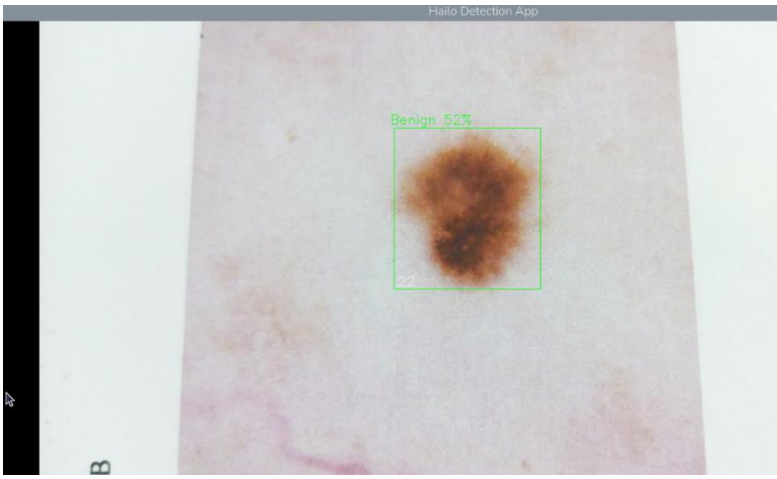
Note the format of the json is given below

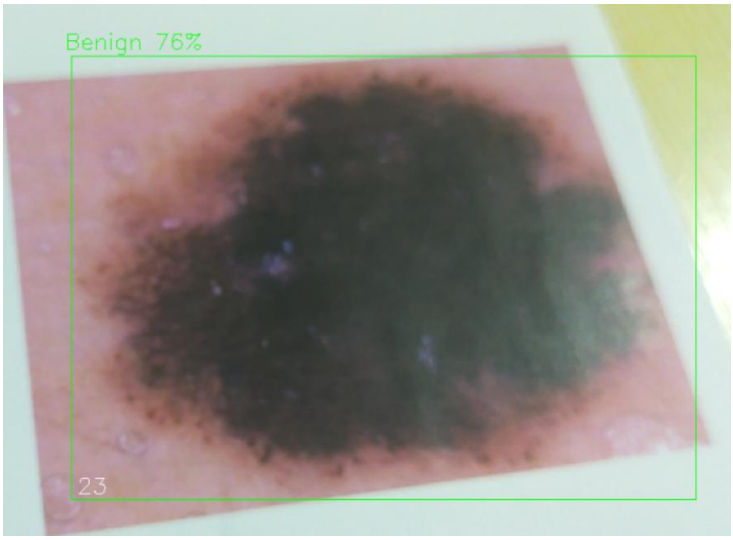
```
{
 "detection_threshold": 0.5,
 "max_boxes": 200,
 "labels": [
  "Unlabled",
  "Benign",
  "Malignant"
 ]
}
```

Normally there will be 2 classes, but in Hailo there will be 3 classes and class 0 will always be unlabelled(or which ever you name).

Outputs

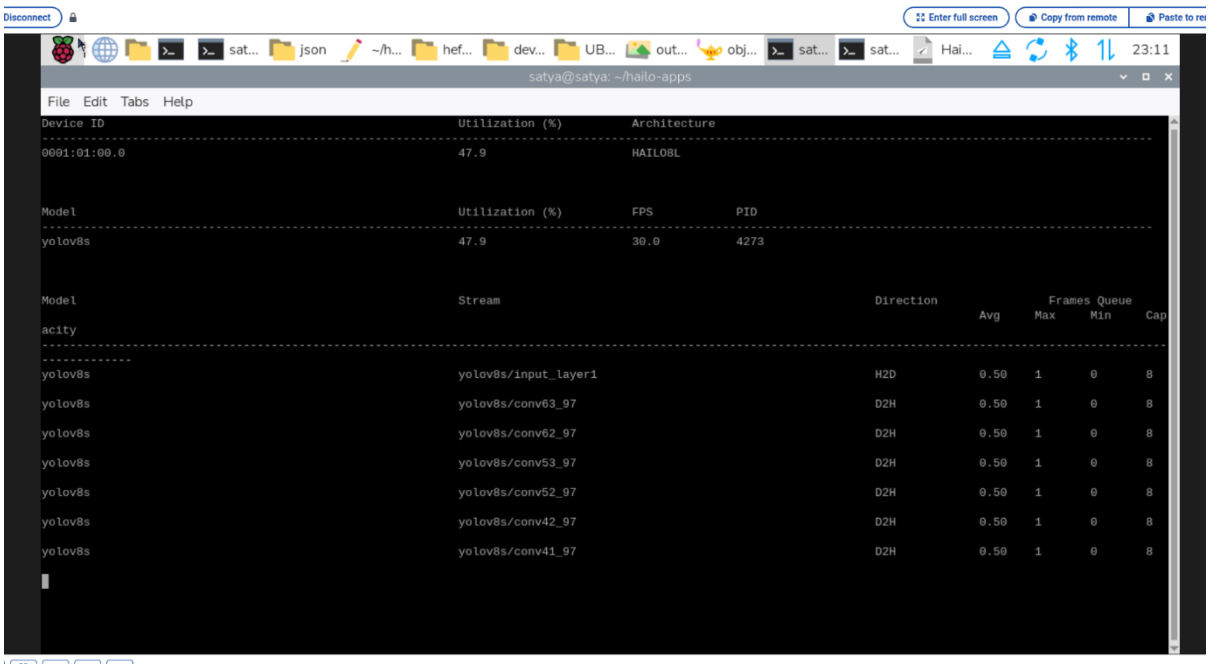


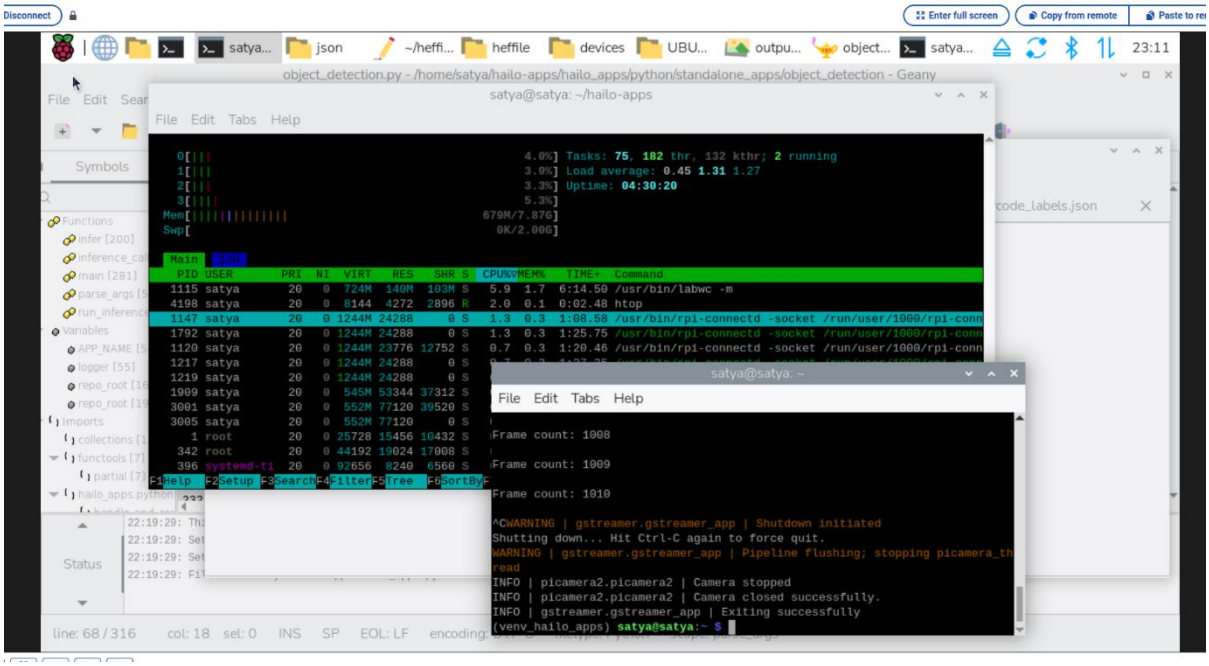


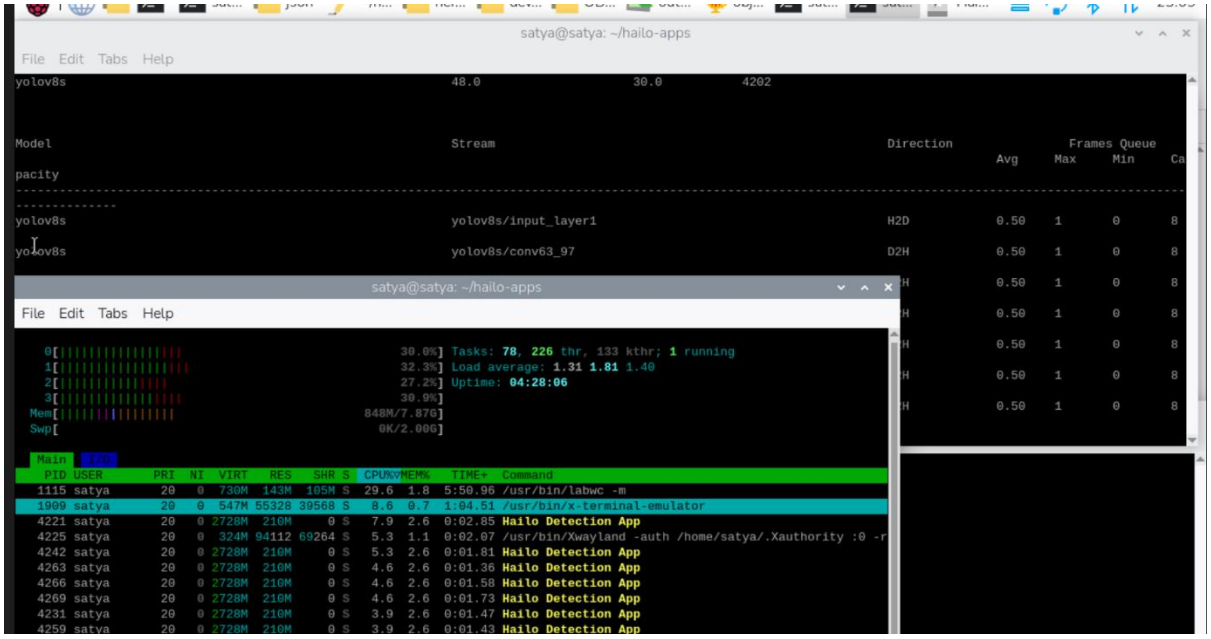


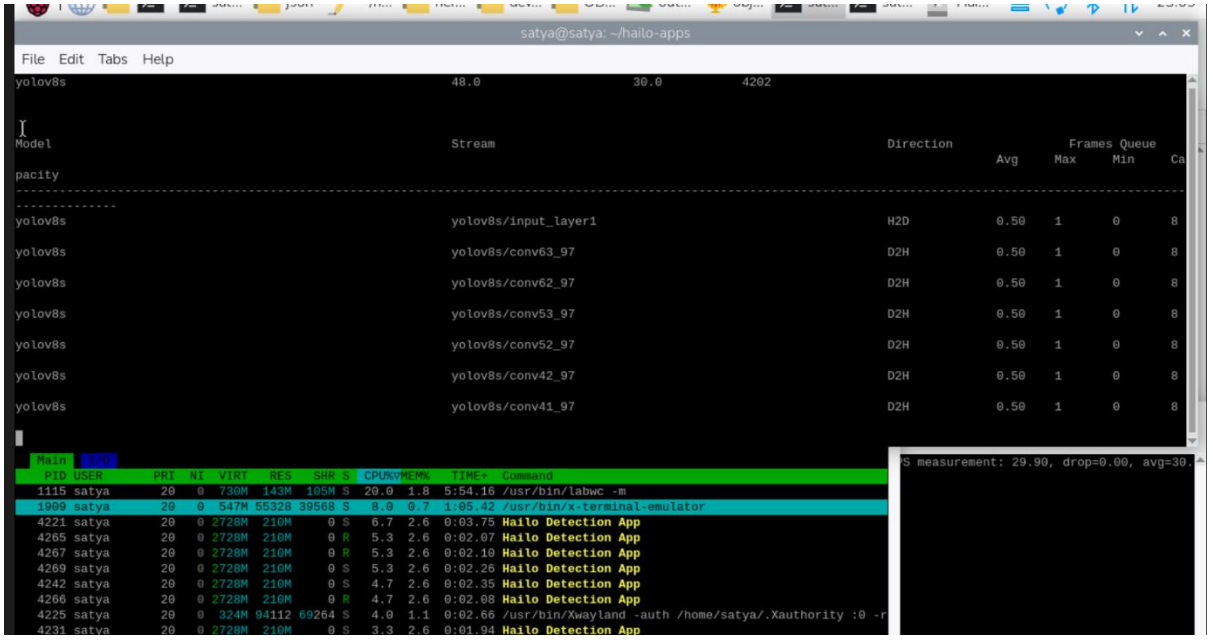
These are the detections using Hailo.

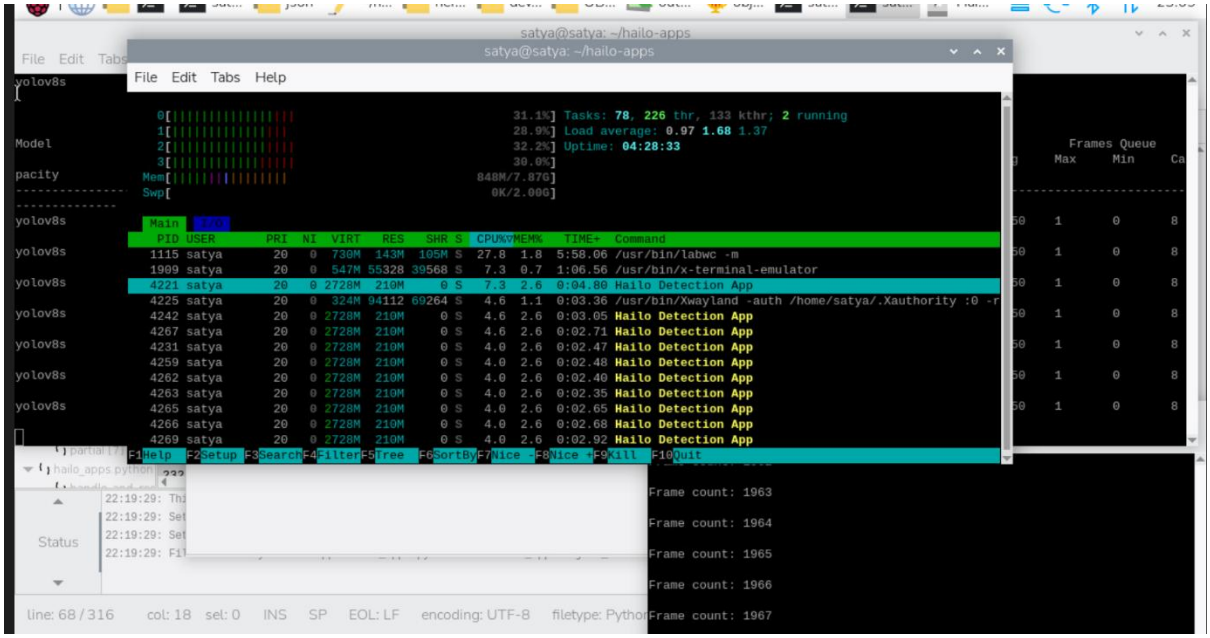
The below are the htop command outputs and the Hailo utilization.

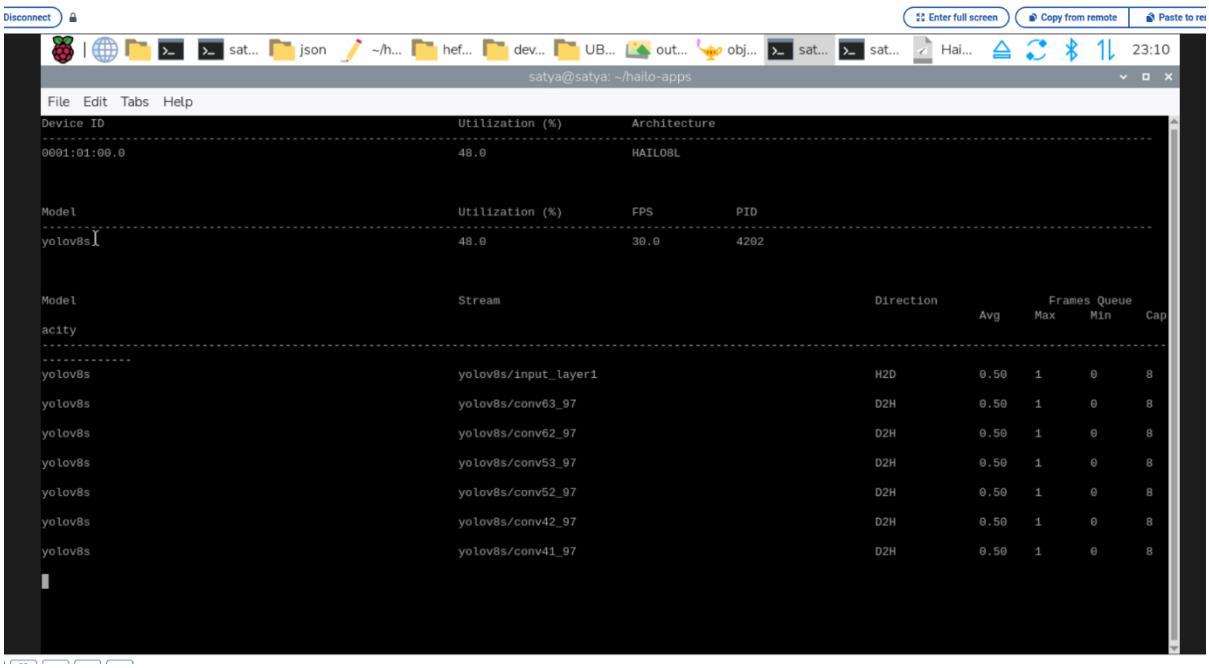

satya@satya: ~/hailo-apps
File Edit Tabs Help
Device ID Utilization (%) Architecture
0001:01:00.0 47.9 HAILO8L
Model Utilization (%) FPS PID
yolov8s 47.9 30.0 4273
Model Stream Direction Frames Queue
Avg Max Min Capacity
yolov8s yolov8s/input_layer1 H2D 0.50 1 0 8
yolov8s yolov8s/conv63_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv62_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv53_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv52_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv42_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv41_97 D2H 0.50 1 0 8

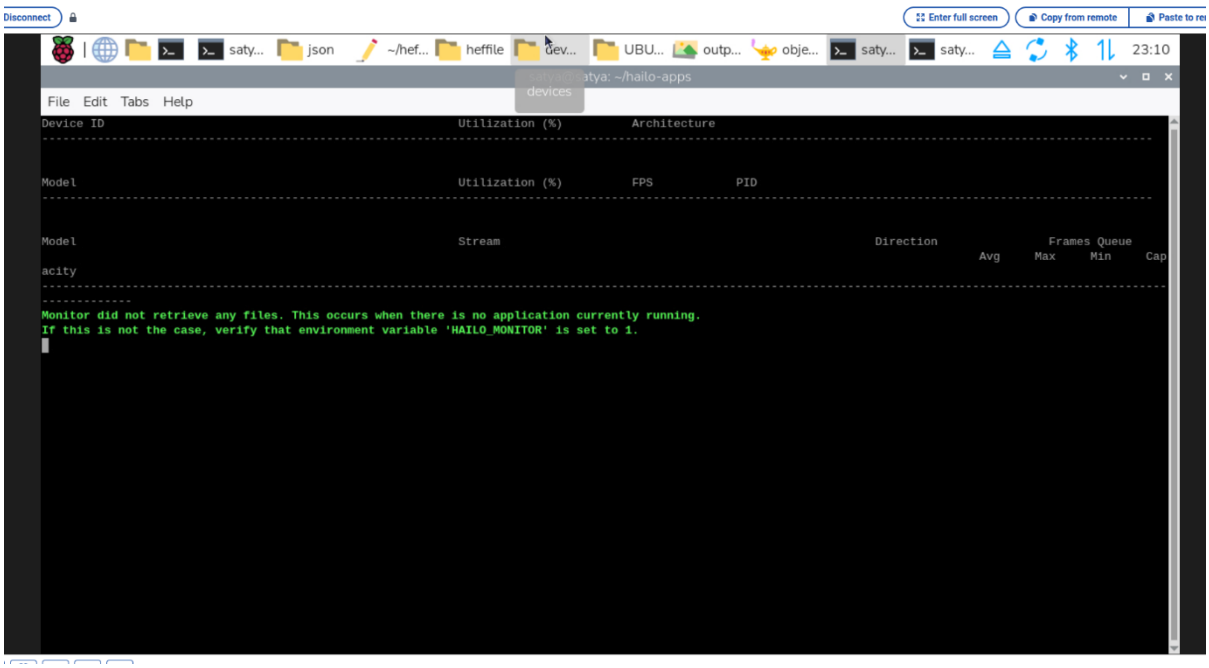

object_detection.py - /home/satya/hailo-apps/hailo_apps/python/standalone_apps/object_detection - Geany
satya@satya: ~/hailo-apps
File Edit Tabs Help
Tasks: 75, 182 thr, 132 kthr; 2 running
Load average: 0.45 1.31 1.27
Uptime: 04:30:20
Mem[670M/7.87G]
Swp[0K/2.00G]
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1115 satya 20 0 724M 140M 103M S 5.9 1.7 6:14.50 /usr/bin/labwc -m
4198 satya 20 0 8144 4272 2896 R 2.0 0.1 0:02.48 htop
1147 satya 20 0 1244M 24288 0 S 1.3 0.3 1:08.58 /usr/bin/rpi-connectd -socket /run/user/1000/rpi-conn
1792 satya 20 0 1244M 24288 0 S 1.3 0.3 1:25.75 /usr/bin/rpi-connectd -socket /run/user/1000/rpi-conn
1120 satya 20 0 1244M 23776 12752 S 0.7 0.3 1:20.46 /usr/bin/rpi-connectd -socket /run/user/1000/rpi-conn
1217 satya 20 0 1244M 24288 0 S
1219 satya 20 0 1244M 24288 0 S
1909 satya 20 0 545M 53344 37312 S
3001 satya 20 0 552M 77120 39520 S
3005 satya 20 0 552M 77120 0 S
1 root 20 0 25728 15456 10432 S
342 root 20 0 44192 19024 17008 S
396 systemd-ti 20 0 92656 8240 6560 S
F1Help F2Setup F3SearchF4FilterF5Tree F6SortByF
satya@satya: ~
File Edit Tabs Help
Frame count: 1008
Frame count: 1009
Frame count: 1010
^CWARNING | gstreamer.gstreamer_app | Shutdown initiated
Shutting down... Hit Ctrl-C again to force quit.
WARNING | gstreamer.gstreamer_app | Pipeline flushing; stopping picamera_th
read
INFO | picamera2.picamera2 | Camera stopped
INFO | picamera2.picamera2 | Camera closed successfully.
INFO | gstreamer.gstreamer_app | Exiting successfully
(venv_hailo_apps) satya@satya:~ $
Symbols
Status
line: 68 / 316 col: 18 sel: 0 INS SP EOL: LF


satya@satya: ~/hailo-apps
File Edit Tabs Help
yolov8s 48.0 30.0 4202
Model Stream Direction Frames Queue
Avg Max Min Ca
pacity
yolov8s yolov8s/input_layer1 H2D 0.50 1 0 8
yolov8s yolov8s/conv63_97 D2H 0.50 1 0 8
satya@satya: ~/hailo-apps
File Edit Tabs Help
0[30.0%] Tasks: 78, 226 thr, 133 kthr; 1 running
1[32.3%] Load average: 1.31 1.81 1.40
2[27.2%] Uptime: 04:28:06
3[30.9%]
Mem[848M/7.87G]
Swp[0K/2.00G]
Main I/O
PID USER PRI NI VIRT RES SHR S CPU%▽MEM% TIME+ Command
1115 satya 20 0 730M 143M 105M S 29.6 1.8 5:50.96 /usr/bin/labwc -m
1909 satya 20 0 547M 55328 39568 S 8.6 0.7 1:04.51 /usr/bin/x-terminal-emulator
4221 satya 20 0 2728M 210M 0 S 7.9 2.6 0:02.85 Hailo Detection App
4225 satya 20 0 324M 94112 69264 S 5.3 1.1 0:02.07 /usr/bin/Xwayland -auth /home/satya/.Xauthority :0 -r
4242 satya 20 0 2728M 210M 0 S 5.3 2.6 0:01.81 Hailo Detection App
4263 satya 20 0 2728M 210M 0 S 4.6 2.6 0:01.36 Hailo Detection App
4266 satya 20 0 2728M 210M 0 S 4.6 2.6 0:01.58 Hailo Detection App
4269 satya 20 0 2728M 210M 0 S 4.6 2.6 0:01.73 Hailo Detection App
4231 satya 20 0 2728M 210M 0 S 3.9 2.6 0:01.47 Hailo Detection App
4259 satya 20 0 2728M 210M 0 S 3.9 2.6 0:01.43 Hailo Detection App


satya@satya: ~/hailo-apps
File Edit Tabs Help
yolov8s 48.0 30.0 4202
Model Stream Direction Frames Queue
Avg Max Min Ca
pacity
yolov8s yolov8s/input_layer1 H2D 0.50 1 0 8
yolov8s yolov8s/conv63_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv62_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv53_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv52_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv42_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv41_97 D2H 0.50 1 0 8
Main I/O
PID USER PRI NI VIRT RES SHR S CPU%▽MEM% TIME+ Command
1115 satya 20 0 730M 143M 105M S 20.0 1.8 5:54.16 /usr/bin/labwc -m
1909 satya 20 0 547M 55328 39568 S 8.0 0.7 1:05.42 /usr/bin/x-terminal-emulator
4221 satya 20 0 2728M 210M 0 S 6.7 2.6 0:03.75 Hailo Detection App
4265 satya 20 0 2728M 210M 0 R 5.3 2.6 0:02.07 Hailo Detection App
4267 satya 20 0 2728M 210M 0 R 5.3 2.6 0:02.10 Hailo Detection App
4269 satya 20 0 2728M 210M 0 S 5.3 2.6 0:02.26 Hailo Detection App
4242 satya 20 0 2728M 210M 0 S 4.7 2.6 0:02.35 Hailo Detection App
4266 satya 20 0 2728M 210M 0 R 4.7 2.6 0:02.08 Hailo Detection App
4225 satya 20 0 324M 94112 69264 S 4.0 1.1 0:02.66 /usr/bin/Xwayland -auth /home/satya/.Xauthority :0 -r
4231 satya 20 0 2728M 210M 0 S 3.3 2.6 0:01.94 Hailo Detection App
S measurement: 29.90, drop=0.00, avg=30.


satya@satya: ~/hailo-apps
File Edit Tabs Help
Tasks: 78, 226 thr, 133 kthr; 2 running
Load average: 0.97 1.68 1.37
Uptime: 04:28:33
Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1115 satya 20 0 730M 143M 105M S 27.8 1.8 5:58.06 /usr/bin/labwc -m
1909 satya 20 0 547M 55328 39568 S 7.3 0.7 1:06.56 /usr/bin/x-terminal-emulator
4221 satya 20 0 2728M 210M 0 S 7.3 2.6 0:04.80 Hailo Detection App
4225 satya 20 0 324M 94112 69264 S 4.6 1.1 0:03.36 /usr/bin/Xwayland -auth /home/satya/.Xauthority :0 -r
4242 satya 20 0 2728M 210M 0 S 4.6 2.6 0:03.05 Hailo Detection App
4267 satya 20 0 2728M 210M 0 S 4.6 2.6 0:02.71 Hailo Detection App
4231 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.47 Hailo Detection App
4259 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.48 Hailo Detection App
4262 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.40 Hailo Detection App
4263 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.35 Hailo Detection App
4265 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.65 Hailo Detection App
4266 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.68 Hailo Detection App
4269 satya 20 0 2728M 210M 0 S 4.0 2.6 0:02.92 Hailo Detection App
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit
Frames Queue
Max Min
Frame count: 1963
Frame count: 1964
Frame count: 1965
Frame count: 1966
Frame count: 1967
Status
line: 68 / 316 col: 18 sel: 0 INS SP EOL: LF encoding: UTF-8


Disconnect
Enter full screen
Copy from remote
satya@satya: ~/hailo-apps
File Edit Tabs Help
Device ID Utilization (%) Architecture
0001:01:00.0 48.0 HAILO8L
Model Utilization (%) FPS PID
yolov8s 48.0 30.0 4202
Model Stream Direction Frames Queue
Avg Max Min Capacity
yolov8s yolov8s/input_layer1 H2D 0.50 1 0 8
yolov8s yolov8s/conv63_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv62_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv53_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv52_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv42_97 D2H 0.50 1 0 8
yolov8s yolov8s/conv41_97 D2H 0.50 1 0 8



The max cpu usage here is about 25% and about 50% of the hailo computation is used here.

We still have the class imbalance problem due to the dataset.

Note.

You would encounter error during the compilation of ONNX to .hef in the following possible ways

- ➔ start node and end node names.
- ➔ Name of the architecture
- ➔ Correct location of the files
- ➔ Correct names in labels.json (add “class 0” as unlabelled)